

When an atmospheric profile hits **extreme winter** (intense cold and dry air) or **extreme summer** (intense heat and high humidity), SARTA's restricted polynomial framework (relying only on linear terms, squares, and square roots) undergoes two distinct tests: **thermodynamic boundaries** and **coefficient training limits**.

1. Behavior in Extreme Winter Limits (The Cold/Dry Test)

In environments like polar regions or the high stratosphere, temperatures drop far below the reference profile, making ΔT a large negative number. Simultaneously, water vapor mass drops to near zero ($W_r \approx 0$).

- **Preventing Negative Optical Depths:** Because SARTA lacks erratic cubic or quartic terms, the quadratic term $(\Delta T)^2$ always yields a stable, predictable positive value. It cannot flip the sign of the overall calculation.
- **The Line Saturation Collapse:** As moisture vanishes, the square root term $\sqrt{W_r}$ drops smoothly toward zero without causing a "divide-by-zero" computational error. The line absorption loop effectively shuts down cleanly, allowing the satellite channels to see straight through to the frozen surface below.

2. Behavior in Extreme Summer Limits (The Hot/Humid Test)

In tropical marine environments, temperatures soar and humidity skyrockets, pushing water vapor levels far above the reference standard ($W_r \gg 1$).

- **The Continuum Explosion:** This is where the squared term (W_r^2) comes into play. Because collision-based absorption scales with the square of the mass, a high summer humidity value causes the continuum optical depth (τ_{cont}) to grow exponentially. SARTA accurately captures the atmospheric "greenhouse blanketing" that completely masks the surface boundary in thermal window channels.
- **Controlled Extrapolation:** Because the polynomial order is kept low (quadratic), if a historic heatwave pushes temperatures slightly outside the model's design specifications, the curve continues along a smooth parabola rather than oscillating wildly.

How the Coefficients are Trained for Boundaries

To ensure that these squares and square roots behave properly at the edges of reality, the static regression coefficients (A_i) must be trained using a diverse, highly specific database of global conditions.

The UMBC Atmospheric Spectroscopy Laboratory trains SARTA using a **48-Profile or 52-Profile Training Set**:

- **Not a Simple Average:** The training database does not just use average global weather. Instead, it deliberately samples the absolute statistical extremes of the planet.
- **The Grid Matrix:** It includes profiles from the Sahara Desert at noon, the Antarctic interior

in winter, tropical super-cell environments, and high-altitude mountain regions.

- **The Tuning Pass:** By running slow, hyper-accurate line-by-line models (LBLRTM) on these 52 extreme profiles across thousands of viewing angles, they generate a diverse dataset. The SARTA regression coefficients are then solved using a least-squares fit against this database.

Because the training set covers the absolute physical boundaries of Earth's atmosphere, and because the predictors are restricted to stable squares and square roots, SARTA can safely process real-time weather data from any spot on the globe without crashing or generating unrealistic brightness temperatures.

Now that you have seen how SARTA's math handles extreme weather boundaries, we have covered its core architecture, inputs, and limitations. Is there any other aspect of fast radiative transfer modeling or satellite data retrieval you would like to explore next?